

CareerPilot AI

Precision Career Intelligence

From career confusion to a clear, evidence-based path toward employment.

A career readiness workspace for students, graduates, and early-career professionals.



Supporting documentation for project submission • September 2026

Live demo: <https://careerpiolet-ai.vercel.app>

Source code: <https://github.com/aminaahsan925/careerpiolet-ai>

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HOW TO READ THIS DOCUMENT

This document is a complete guide to the deployed prototype. It is organized around the actual product experience so a reader can understand what the student sees, what the system stores, how AI is used, and what a judge can verify.

1. Executive Summary

CareerPilot AI helps students understand why they are unemployed and what they should do to become job-ready. Instead of offering generic courses or motivational advice, the platform compares a student's current profile with the expectations of a selected role and company. It turns identified gaps into a practical roadmap with daily learning, portfolio work, assessments, interview preparation, and recruiter-style feedback.

The project focuses on the gap between education and employability. Students may have degrees, certificates, and theoretical knowledge, but employers look for evidence: real projects, problem-solving, deployment, communication, and the ability to explain technical decisions. CareerPilot connects market reality, personal diagnosis, execution, and proof in one accessible workspace.

CORE PROMISE

Where am I now? Why am I being rejected? What does my target company expect? What should I learn, build, and practice next?

The prototype is deployed and end-to-end usable: a judge can register, complete onboarding, run a diagnosis, inspect market reality for a target role and company, generate a personalized roadmap, complete daily work, practice interviews, and receive recruiter-style feedback in a single session.

2. Problem, Audience, and Context

Students in Pakistan and other emerging markets often learn without a clear target. They do not know which skills matter for a specific job, why applications fail, what projects employers value, or how to prepare for interviews. Students outside major cities may also lack access to mentors, professional networks, and expensive career programs.

The result is a costly mismatch: months spent learning skills the market no longer asks for, resumes that never pass automated screening, and interviews that end at the first technical question. Existing tools address single fragments of this problem — job boards list openings, course platforms sell content, resume builders format text — but none of them connect a student's actual evidence to what a specific employer will actually screen for.

WHO IT IS FOR

- University students choosing a direction before graduation
- Recent graduates facing rejection without feedback
- Self-taught developers proving skills without a CS degree
- Career switchers entering tech from other fields
- Universities and training organizations seeking a structured employability journey

For institutions, CareerPilot offers a repeatable framework: every learner's readiness, gaps, and progress become measurable instead of anecdotal. This makes the platform relevant beyond individual use and into employability programs at scale.

3. Solution and User Journey

CareerPilot converts uncertainty into a sequence of decisions and actions:

01 Understand

Onboarding captures education, experience, skills, projects, certifications, and career goals.

02 Diagnose

The platform identifies missing skills, weak proof, outdated knowledge, and rejection risks.

03 Compare

Market Reality connects the student with role, company, Pakistan, and global hiring expectations.

04 Execute

A personalized roadmap provides daily learning, practical tasks, and portfolio-focused projects.

05 Practice

MCQs, interview preparation, and recruiter simulation turn knowledge into confident evidence.

06 Progress

Assessments, notifications, readiness, and completion tracking make progress visible.

Every stage feeds the next. The diagnosis defines the gap, the market reality defines the target, the roadmap defines the work, and assessments plus recruiter feedback verify that the work produced believable proof. No stage treats the student as a blank slate.

4. Onboarding and Career Profile

CareerPilot begins by building a structured profile instead of asking the student to start with a generic chatbot question. The onboarding flow is a guided, nine-step setup that captures the learner's starting point, constraints, evidence, and target direction.

About You

Name, current status, and the learner's present situation establish the context used across the workspace.

Education

Education level, degree, university, academic year, and graduation timing help the system understand the learner's stage.

Experience

The student records academic work, personal projects, freelance work, internships, professional work, and open-source contributions.

Projects

Each project can include a name, description, technologies, project type, and public URL. This becomes evidence that later modules can inspect.

Skills

Students select skills from the catalog, add custom skills, and describe confidence as beginner, comfortable, or strong.

Career Direction

Learning history, career clarity, certifications, and target role help separate an uncertain learner from someone ready for focused progression.

Situation and Constraints

The student identifies the biggest blocker, how well education prepared them, and the weekly time available for learning.

WHY THIS MATTERS

The profile is the shared context behind diagnosis, market research, roadmap generation, resume matching, mentor guidance, and recruiter feedback.

5. Dashboard and Career State

The dashboard is the student's career command center. It brings readiness signals, the next action, roadmap status, applications, skills, market context, and mentor access into one view. It is designed to answer one practical question: what should I do next?

DASHBOARD AREA	PURPOSE
Readiness cards	Show career, resume, interview, and evidence signals when the student has completed the relevant analysis.
Career diagnosis	Surfaces career stage, the main blocker, and the next best action.
Recommended roadmap	Shows current learning stage, progress, and upcoming work.
Applications	Tracks companies, roles, application status, interview dates, notes, and outcomes.
Top skills	Keeps the student's recorded skill inventory visible and connected to resume evidence.
Market Reality	Links the student's target role to employer expectations and demand signals.
Today's Plan	Provides a focused daily action rather than an open-ended list of courses.
AI Resume Review	Shows whether the resume has been analyzed and directs the student to the next improvement.

The dashboard reads from a single career state assembled on the server. That state combines profile, goal, skills, projects, resume analysis, diagnosis, readiness, roadmap progress, and target context. This prevents each page from giving advice from a different version of the student.

6. Career Diagnosis and Readiness

Diagnosis is the product's central reasoning feature. It audits the student's profile against a selected role and company, then explains what is missing, why the gap matters, and what evidence should be produced next.

The diagnosis wizard has three explicit decisions: select the target employer, select the target position, and launch the rejection diagnosis. A custom company can be entered when the employer is not in the featured list. The system can also match a company truth profile to known employer expectations.

Readiness score and breakdown

The system stores an overall readiness value, career stage, and a breakdown of the factors that influence the student's current position.

Strengths

The report records genuine strengths found in the student's saved profile and evidence. The diagnostic instructions explicitly avoid inventing strengths.

Blockers and rejection risks

Each blocker includes the problem, supporting evidence, why it matters from a hiring perspective, and a fix that the student can act on.

Priorities

The report ranks the most important work and connects each priority to the evidence the student should produce.

Next best action

The student receives one immediate action with a reason and a definition of the proof that would make the action credible.

Recovery sequence

The result is ordered into now, next, and after stages so the student can work through a realistic progression.

DIAGNOSTIC PRINCIPLE

A listed skill is a claim. A deployed project, measurable outcome, or explained technical decision is evidence.

7. Market Reality and Future Tech

Market Reality gives the student a reference point outside their own assumptions. It connects the target role to employer expectations, demand signals, salary context, skill gaps, and future technology trends. The page supports global research and optional Pakistan-focused context.

01 Role signal

Identify the skills, responsibilities, and proof expected for the selected role.

02 Company signal

Use the target company and researched employer context to make the guidance more specific.

03 Local lens

Review Pakistan-focused opportunities and salary context alongside remote and global paths.

04 Demand signal

Separate established skills from high-growth areas and technologies that deserve monitoring.

05 Source layer

Show research sources and summaries so the student can inspect where market claims came from.

06 Action layer

Save a technology to a learning track or add a market-relevant skill to the roadmap.

Future Tech is a guided research experience rather than a list of buzzwords. Each technology can be explored through four questions: what it is, why it matters, whether the student should learn it, and how to learn it. The detail view can include prerequisites, use cases, career relevance, confidence, a learning path, a first project, sources, and personal tracking status.

8. Roadmap and Daily Execution

The roadmap converts diagnosis into a sequence of learning stages. It is personalized from the student's profile, target role, evidence, available weekly time, and market context. The product treats learning as work that should produce visible proof.

Learning stage

Each stage has a title, explanation, expected outcome, and practical context so the student understands why it exists.

Daily task

A roadmap day can include an explanation, a concrete task, a project direction, and a definition of done.

Skill and evidence link

The work is connected to the skill gap and the evidence the student needs for the target role.

Completion tracking

Students can mark practical work complete. Progress remains available after navigation and refresh because it is persisted.

Assessment checkpoint

Role-specific MCQs test understanding and return explanations instead of only showing a score.

Certificate

When the learning path is complete, the product can display a CareerPilot completion certificate for the selected target role.

This design keeps CareerPilot focused on a small number of meaningful actions. The student does not only consume recommendations. The student completes work, records progress, and builds a portfolio trail.

9. Flight Plan and Job Readiness

Flight Plan is the job-description readiness workflow. It translates a specific vacancy into a practical comparison between what the role asks for and what the student's saved profile can prove.

FLIGHT PLAN STAGE	WHAT THE STUDENT RECEIVES
Target role selection	A selected or entered role becomes the assessment target.
Job description input	The system receives the role requirements that matter for this application.
Skill mirror	Required skills are compared with the student's recorded skills and evidence.
Rejection truth	The product explains the most likely reasons the application could fail screening.
Market truth note	The student sees how market expectations affect the readiness judgment.
Fit verdict	The result summarizes current fit and the gap that must be closed.
Action plan	The student gets practical next steps instead of a generic fit percentage.

PRODUCT VALUE

Flight Plan turns a job description into a preparation plan before the student spends time applying.

10. Resume Intelligence and ATS

Resume Intelligence is a complete upload, parsing, scoring, and positioning workflow. Students can upload a PDF, DOCX, or TXT resume up to 5 MB. The interface supports file selection and drag-and-drop, then stores the active file and latest analysis.

RESUME OUTPUT	WHAT IT EXPLAINS
ATS Score	Keyword and format fit based on structural checks and target-role alignment.
Impact Score	Use of action verbs, measurable outcomes, and stronger accomplishment framing.
Role Match	Compatibility between the resume and the student's target role.
Executive Summary	A recruiter-style summary of the current presentation.
Competitive Strengths	The strongest signals already present in the resume.
ATS Red Flags and Gaps	Missing details, weak phrasing, structure problems, or evidence gaps.
Positioning Rewrites	High-impact recommendations with an explanation of why each change matters.
Extracted Skills	Technical skills detected from the resume text.
Market Target Compatibility	Role matches compared with researched industry roles.

The analysis is grounded in the student's saved career goal and is synchronized back into the shared career state. Resume evidence can therefore influence diagnosis, readiness, recruiter audit, mentor context, and dashboard scores.

RESUME POLICY

CareerPilot treats a resume as evidence that must support the target role, not as a document that only needs attractive formatting.

11. Interview Preparation and Job Mirror

Interview Prep turns the selected role into a focused practice plan. It gives the student a role-specific interview intelligence view instead of a generic list of questions.

Answer Lab

Students practice responses and refine how they explain decisions, projects, trade-offs, and outcomes.

Evidence to bring

The product identifies the projects, metrics, links, and examples that should support the student's claims in an interview.

Expected interview stages

The student can prepare for technical screening, project discussion, behavioral questions, and company-specific evaluation.

Job Mirror

The role is reflected back as an employer expectation model, showing what the student should know and what proof would make the claim believable.

The value of this module is alignment. Interview practice, resume positioning, roadmap work, and recruiter feedback all point toward the same target role and evidence base.

12. Recruiter Audit

Recruiter Audit simulates a hiring review of the student's profile. It is built around the question a recruiter or technical reviewer would ask: is there enough credible evidence to move this candidate to the next stage?

Candidate snapshot

The audit uses the target role, target company, profile, skills, projects, resume analysis, diagnosis, and readiness context.

Recruiter perspective

The report identifies red flags, missing production proof, unclear positioning, and risks that may prevent an interview invitation.

Evidence review

Projects and resume claims are inspected as proof. Public URLs and measurable outcomes carry more weight than unverified lists of tools.

Persistent recruiter session

The recruiter conversation is stored so the student can return to the same audit context and continue improving the profile.

Pipeline context

The interface presents the review as a hiring pipeline with clear movement from profile audit to interview readiness.

RECRUITER TEST

If a student says they know a technology, the audit asks what they built, deployed, measured, and can explain.

13. AI Mentor and Progress System

The AI Mentor is context-aware. It does not start every conversation from zero. The server builds a career context from the student's profile, goals, skills, projects, resume analysis, diagnosis, readiness, and roadmap progress before generating a response.

- Suggested questions help students start with common career decisions.
- Career insights summarize the current state and expose the next useful question.
- Recommended actions connect mentor advice to roadmap, resume, diagnosis, or interview work.
- Chat history is stored so the student can return to an ongoing coaching conversation.
- The mentor is designed for direct, specific guidance rather than generic encouragement.

The progress system completes the loop. Notifications can surface roadmap progress, assessment events, and career nudges. The dashboard summarizes the same progress, and the certificate provides a visible completion outcome when the path is finished.

Profile remains editable after onboarding. Students can update basic information, knowledge, skills, avatar, projects, and account details as their evidence improves.

14. Technical Architecture and Security

CareerPilot uses a React and TypeScript client with TanStack Router and TanStack Query. Reusable UI components, server functions, loading states, error handling, and cached queries keep the product maintainable as the feature set grows.

LAYER	IMPLEMENTATION
Client	React 19, TypeScript, TanStack Router, TanStack Query, Tailwind CSS, Motion, and reusable UI components.
Server functions	Authenticated functions validate input, load user-scoped state, call AI or research providers, and persist results.
Database	Supabase PostgreSQL stores profiles, goals, skills, projects, resumes, diagnoses, roadmaps, assessments, applications, notifications, and recruiter sessions.
Storage	Supabase Storage supports user files such as resumes and avatars with user-scoped access.
AI providers	Groq, Gemini, and OpenRouter can be selected through server-side configuration with fallback handling.
Research	Tavily-backed research supports market and technology intelligence with cached results and source links.
Deployment	Vercel provides the serverless deployment target for the working prototype.

Credential boundary

Privileged AI and research keys remain on the server. The browser receives only the data needed for the authenticated experience.

User isolation

Row Level Security and user-scoped queries protect personal profiles, resumes, projects, progress, and conversations.

Validation and fallback

Server validation protects assessment and profile operations. Provider fallback and cached results keep core workflows useful when an external service is unavailable.

15. Data Model and Persistence

The product is persistent by design. Important actions create or update records instead of living only in browser state. This is what makes the experience resume-able after refresh and allows multiple modules to share the same career context.

DATA AREA	STORED INFORMATION
Identity and profile	Authentication identity, name, education, status, university, goals, weekly time, and career context.
Skills	Skill catalog selections, custom skills, confidence, and evidence strength.
Projects	Project name, description, technologies, URL, project type, and evidence metadata.
Career target	Target role, target industry, target job, and company context.
Diagnosis	Readiness, stage, strengths, blockers, priorities, next best action, sequence, and market benchmark.
Resume	Uploaded file metadata, parsed analysis, scores, skills, weaknesses, recommendations, and role matches.
Roadmap	Generated stages, days, task content, completion, assessments, attempts, and certificate status.
Engagement	Applications, notifications, mentor chat messages, recruiter sessions, and future-tech tracking.

The database migration history shows the product evolving through added diagnostic intakes, market reality caching, user projects, technology tracking, roadmap versioning, resume storage fixes, and recruiter sessions. That history reflects an iterative prototype rather than a static mockup.

16. Demonstration and Evaluation Guide

A strong demonstration should show the relationship between modules, not only open isolated pages. The following sequence gives a judge the clearest view of the product's value.

- 1 Profile**
Open or create a student account and complete the Know Me onboarding flow.
- 2 Target**
Select a target role and company, or enter a custom company.
- 3 Diagnosis**
Run the company-specific diagnosis and review blockers, priorities, and next action.
- 4 Market**
Open Market Reality to inspect role expectations, demand, local context, and sources.
- 5 Resume**
Upload a PDF, DOCX, or TXT resume and review ATS, impact, and role-match scores.
- 6 Plan**
Generate the personalized roadmap and open the first practical day.
- 7 Proof**
Complete a task, take the MCQ, and inspect the explanation and saved progress.
- 8 Interview**
Open Job Mirror and practice a role-specific answer with the evidence to bring.
- 9 Recruiter**
Run Recruiter Audit to see how the same evidence looks from an employer's perspective.
- 10 Mentor**
Ask the AI Mentor for the next action and show that it remembers the student's context.

WHAT THE JUDGE SHOULD REMEMBER

CareerPilot carries one target and one evidence base through the entire product journey. Every module answers the question raised by the previous module.

17. Impact, Feasibility, and Next Steps

CareerPilot addresses a practical challenge for Pakistan and other emerging markets: educated people often need clearer information, stronger proof, and more structured support before they can compete for work. The product turns that challenge into a measurable personal workflow.

Immediate value

Students can see the distance between their current evidence and the expectations of a target role instead of relying on guesswork.

Accessible guidance

The AI Mentor, diagnosis, and roadmap make context-aware guidance available without requiring every student to find a private mentor.

Low-cost foundation

Serverless deployment, cached research, provider fallback, and Supabase persistence create a practical foundation for a hackathon-scale product.

Institutional potential

Universities and training programs could use the same structure to track learner readiness, evidence, activity, and outcomes.

Next product steps

Expand role and company coverage, improve market source quality, add deeper interview banks, support institutional cohorts, and measure the relationship between roadmap completion and job outcomes.

PROJECT SUMMARY

CareerPilot AI connects personal context, market reality, practical learning, and employer proof in one deployed career readiness workspace.

PROJECT LINKS

Live demo	https://careerpiolet-ai.vercel.app
Source code	https://github.com/aminaahsan925/careerpiolet-ai